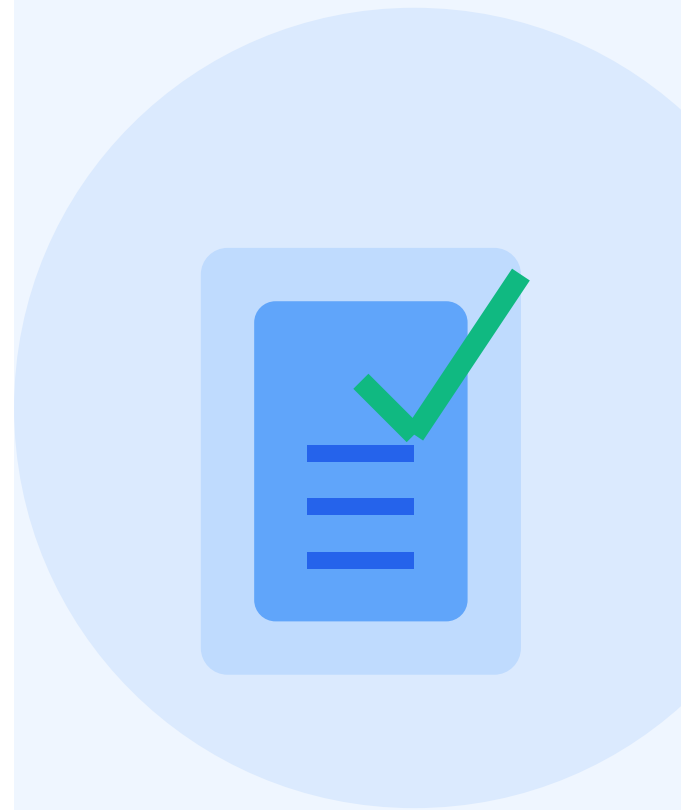


Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval



Report Summary

Report Type:	Business Report
Generated On:	2026-10-03 21:27
Total Records:	102
Total Columns:	21
Target Column:	Loan_Status
Active Model:	Logistic Regression

✓ Data Analysis & Insights

✓ Risk Prediction

✓ Approval Decision

✓ Better Financial Decisions

1. Dataset Overview

Total Records

102

Total Columns

21

Numeric Cols

11

Categorical Cols

10

2. Data Quality

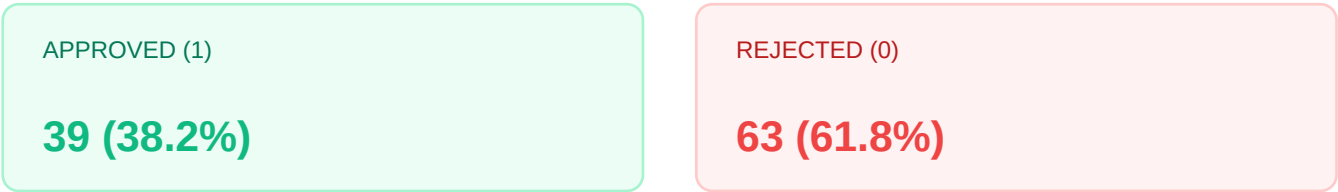
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

✓ Dataset is ready for analysis and model training.

3. Missing Values by Column

✓ No missing values detected.
All columns have complete data.

4. Target Analysis (Loan_Status)



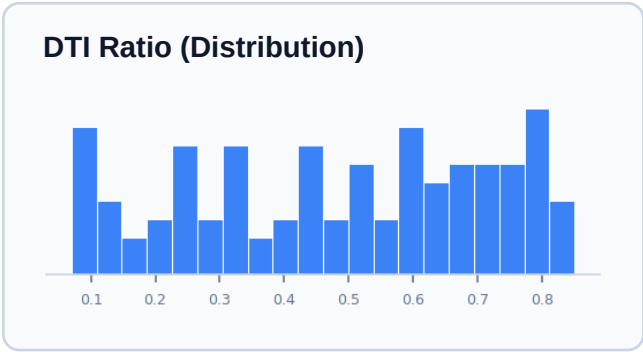
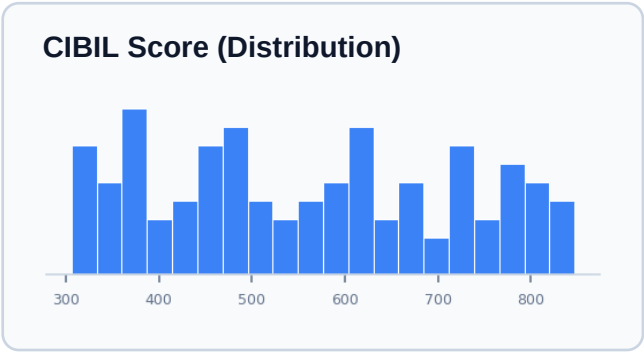
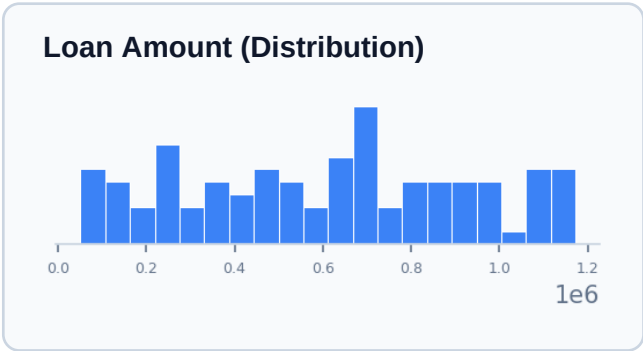
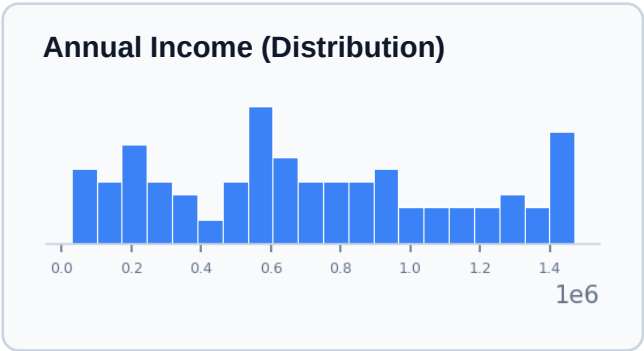
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.8	43.0	21.0	65.0	13.3
Employment_Years	16.1	18.0	0.0	30.0	9.1
Annual_Income	709,578.06	669,646.00	28,267.00	1,471,403.00	422,109.94
Coapplicant_Income	176,856.75	179,208.00	1,542.00	394,540.00	108,531.35
Loan_Amount	608,104.39	638,579.00	51,062.00	1,172,262.00	319,758.03
Loan_Term_Years	12.5	10.0	2.0	30.0	9.0
Cibil_Score	559.5	553.0	306.0	847.0	158.2
Existing_Loans	2.6	3.0	0.0	5.0	1.7
DTI_Ratio	0.5	0.5	0.1	0.8	0.2
Residential_Assets	2,331,195.13	2,215,685.00	11,303.00	4,988,023.00	1,515,562.25
Bank_Assets	1,034,718.12	1,080,174.00	42,534.00	1,971,164.00	551,003.19

6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %	Severity
No significant outliers	-	-	-

7. Distribution of Key Features



8. Model Performance

Accuracy

76.7%

Precision

78.6%

Recall

85.4%

F1 Score

81.9%

ROC-AUC

84.1%

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

9. Active Model Details

Model Name

Logistic Regression

Model Version

v20261003212640

Model Type

Loan Risk Classification

Training Status

Ready for Prediction

10. New Applicant Prediction

Applicant Information		Prediction Result	
Age	56.0	LOW RISK	
Employment Years	27.0	Prediction Confidence: 90.3%	
Annual Income	509000.0	Approval Probability: 90.3%	
Coapplicant Income	52000.0	Risk Probability: 9.7%	
Loan Amount	579000.0	Decision: APPROVE	
Loan Term Years	3.0		
Cibil Score	759.0		
Existing Loans	2.0		
DTI Ratio	0.671		
Residential Assets	1894000.0		
Bank Assets	228000.0		
Gender	Male		
Married	Yes		
Dependents	0		

11. Final Summary

The system analyzed 102 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset.

Total Records 102	Total Columns 21	Selected Model Logistic Regression
Model Accuracy 76.7%	Latest Applicant Result LOW RISK	Approval Probability 90.3%